Suggested 15-species panel for cancer resistance

#	Species	Key Cancer-Resistance Mechanism	Lineage
1	Naked mole-rat (*Heterocephalus glaber*)	High-MW hyaluronan, p16/p27 dual checkpoint, hypoxia tolerance	Rodents
2	Blind mole-rat (*Spalax ehrenbergi*)	IFN-β mediated concerted cell death, p53/Rb pathway divergence	Rodents
3	Damaraland mole-rat (*Fukomys damarensis*)	Contact inhibition, distinct from naked mole-rat — independent mechanism	Rodents
4	Beaver (*Castor canadensis*)	Large rodent defying Peto's Paradox; understudied but genome available	Rodents
5	Bowhead whale (*Balaena mysticetus*)	PCNA/DNA repair divergence, altered cell proliferation kinetics	Cetaceans
6	Sperm whale (*Physeter macrocephalus*)	Largest toothed predator; extreme Peto's Paradox case; TP53 duplications reported	Cetaceans
7	African elephant (*Loxodonta africana*)	20 TP53 copies, LIF6 pro-apoptotic retrogene, enhanced DNA damage response	Proboscideans
8	Asian elephant (*Elephas maximus*)	Independent validation of elephant TP53 story — same lineage, confirms mechanism	Proboscideans
9	Greenland shark (*Somniosus microcephalus*)	Extremely low metabolic rate, minimal cancer reports, genome available	Sharks
10	Little skate (*Leucoraja erinacea*)	Elasmobranch outgroup to shark; comparative anchor for shark-specific signals	Sharks
11	Painted turtle (*Chrysemys picta*)	Negligible senescence, extremely low cancer incidence, good genome assembly	Reptiles
12	Ocean quahog clam (*Arctica islandica*)	500+ year lifespan, exceptional oxidative stress resistance, cancer extremely rare	Molluscs
13	Little brown bat (*Myotis lucifugus*)	DNA damage tolerance, longevity outlier vs body mass, interferon pathway divergence	Bats
14	Elephant shark (*Callorhinchus milii*)	Oldest jawed vertebrate genome; slow evolution rate; baseline for shark comparison	Cartilaginous fish
15	Hydra (*Hydra vulgaris*)	Effectively non-senescent; continuous stem cell renewal; tumour suppression without adaptive immunity	Cnidarians